

Customer Lifetime Value Analysis

An End-to-End CLV Framework: RFM Segmentation - Descriptive CLV - BG/NBD Prediction - Actionable Strategy

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THE FIVE MAIN QUESTIONS

1. Which customers generate the highest lifetime value?
2. Which customers are at risk of churn?
3. How should marketing budgets be allocated across segments?
4. What customer retention strategies should be deployed?
5. Which customers deserve priority in loyalty programs?

DATASET AT A GLANCE

62,767

Transactions

2023–2025

8,510

Unique Customers

3-yr cohort

\$3.85

Mean CLV / month

full base

\$102.61

Mean Pred. CLV

52-wk horizon

METHODOLOGY & RATIONALE

1. Data Ingestion

The dataset was assembled by joining six PostgreSQL tables (ORDER, CUSTOMER, ORDER_ITEM, PRODUCT, CATEGORY, and PRICE_CHANGE) yielding 62,767 transactional records across 2023–2025, addressing 15 total variables, with a stricted procedure to ensure consistency and reliability.

2. Data Quality

- Only transactions with a confirmed price record are retained, eliminating orphaned rows without explicit imputation.
- NaN values were replaced with None before loading into PostgreSQL to guarantee clean column typing throughout the pipeline
- **Zero missing values** are present in the dataset.

3. Data Preparation

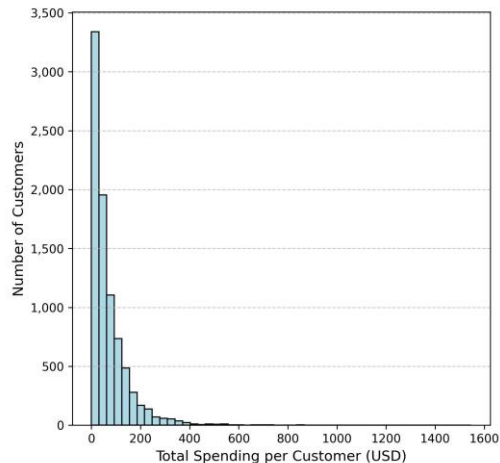
- Fixed change of the **currency** of 83.0 INR/USD on all the numeric variables
- 2 new variables were created *gross_revenue_usd* and *net_revenue_usd*
- **Item_discount** presented anomalous patterns, suggesting that these discounts were recorded at an order level, and mechanically distributed across individual items, making its aggregation unreliable
- A floor of **zero** is applied to *net_revenue_usd* using .clip(lower=0) to prevent negative revenue entries from distorting the Monetary signal
- **New dataset** created (*df_orders*) created using order-level aggregated revenue and used for the following steps of RFM, CLV, BG/NBD modelling

DATASET SCHEMA - DF_ITEMS

Column	Type	Role
order_id, product_id	String	Used for the merge of the datasets
customer_id	String	Unit of analysis
quantity, item_discount, unit_price	Numeric	Revenue inputs
brand, category_name, city, state, segment	Categorical	EDA & segmentation
order_date, signup_date	Date	Temporal anchors
year, week	Numeric	Temporal coordinates for the JOIN of PRICE_CHANGE
gross_revenue_usd	Numeric	quantity x unit_price_usd
net_revenue_usd	Numeric	gross_revenue_usd – item_discount_usd

2. Exploratory Data Analysis

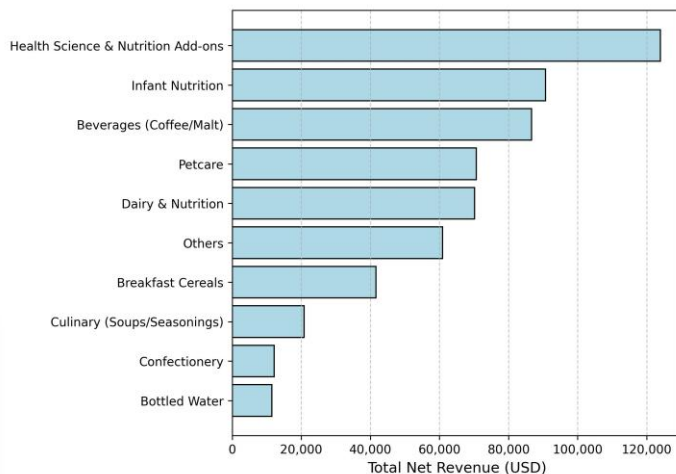
1. Customer Spending Distribution



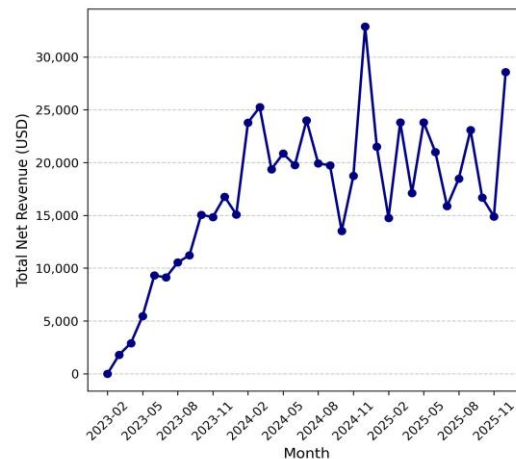
Customer level aggregation is strongly **right-skewed**. Most customers concentrate in a low-frequency, low-spend bracket, while a thin tail of highly active customers generates a disproportionate share of total revenue

Net revenue was aggregated at the `category_name` level from `df_items`. The top categories by total revenue are Nutrition Add-ons, Infant Nutrition and Beverages (Coffee/Malt).

3. Revenue by Category



6. Time-based Purchasing Trends



Monthly net revenue was computed from `df_orders` over the full 36-month observation window (January 2023 – December 2025). This is important for interpreting the BG/NBD calibration/holdout split used in Part E, where 2023–2024 serves as the training window and 2025 as the holdout.

3. RFM Analytical Framework & CLV Modeling Methodology

RFM ANALYSIS METHODOLOGY

RFM scores (1–5) were assigned via **quantile-based** binning to handle duplicate bin edges and replace fragile rank-based workarounds.

Recency scoring is intentionally **inverted** (low recency → high score), reflecting the business logic that recently active customers are more valuable, a deliberate asymmetry in the scoring function.

Of 125 theoretically possible RFM segments, 111 were observed; 83 contained fewer than 100 customers and were deemed statistically insufficient.

So, these sparse-segment customers were reassigned via constrained **3D Euclidean distance**, with a hard constraint requiring the target segment to share the same R (Recency) score, preserving the most business-critical dimension, reducing to only 28 segments, with 2,985 customers (35.1%) successfully reassigned from sparse to representative segments.

CLV DESCRIPTIVE

Descriptive CLV used a customer-level three-component model:

$CLV = (APV \times PF) / \text{Churn Rate}$, computed **individually** per customer rather than as dataset-wide aggregates.

Purchase Frequency (PF) is **normalized** by each customer's active lifetime in months, capturing individual purchase rates while accounting for tenure heterogeneity.

The results were computed for all 8,510 customers.

The full-base mean CLV of \$3.85/month rises to \$6.11/month when restricted to the 3,154 **active customers** (Recency $\leq$ 90 days), confirming that active customers generate 59% more monthly value on average.

The **top 20** customers by CLV (led by CUST002744 at \$78.90/month) share a consistent behavioural profile: from 39 to 152 orders, APV of \$9-16, and tenures spanning the full three-year window, confirming habitual purchase behaviour rather than one-off spending spikes.

DESCRIPTIVE CLV

$$CLV = (APV \times PF) \div \text{Churn Rate}$$

Customer-level, monthly \$

CHURN DEFINITION

Recency > 90 days

62.9% of base classified churned

ACTIVE CUSTOMER ONLY

Mean CLV = \$6.11/month

Against \$3.85/month for the full dataset

4. Predictive CLV& Empirical Evidence

PREDICTIVE CLV: BG/NBD + GAMMA-GAMMA

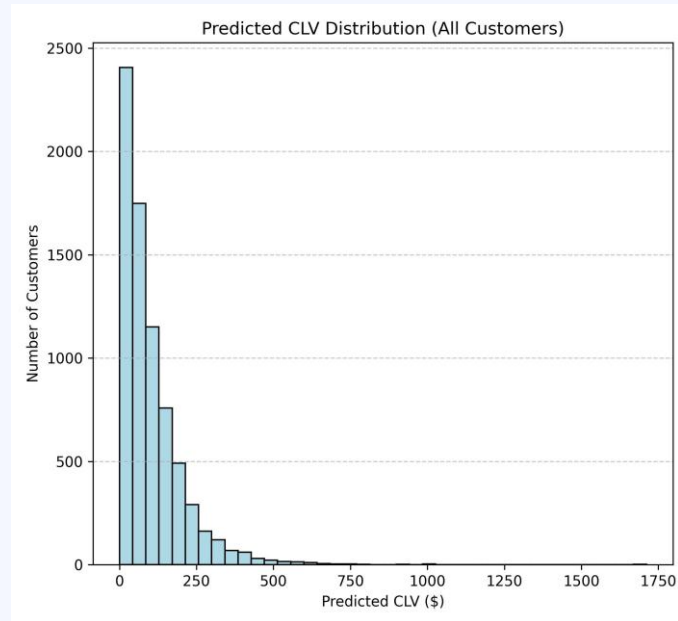
The BG/NBD combined with the Gamma-Gamma model was selected for 3 main reasons: the dataset is not subscription-based, there are enough data history to estimate the parameters of the model and the customers show heterogeneous purchasing behavior.

The data was transformed to be handled by BG/NBD summary in a weekly format. It consists of two separate models:

- BetaGeoFitter models the purchase frequency and dropout process simultaneously.
- GammaGammaFitter models the monetary value of transactions, assuming each customer has an individual mean spend drawn from a Gamma distribution.

The two models were combined to predict individual CLV over a 52-week horizon (after 104 of training) using a monthly discount rate of 1%, reflecting the time value of money.

The predictive model covered 7,369 customers (86.6% of base; 1,141 excluded due to zero monetary value causing log-likelihood instability). All top 20 customers exhibit high recency relative to T (recency $\approx$ T), confirming they remained active throughout the observation window and are likely to continue purchasing in the prediction horizon. The distribution of the CLV prediction and the evaluation metrics are displayed on the right.

**2.345****RMSE**

Holdout period

1.783**MAE**

Holdout period

0.426**R²**

Holdout period

5. Customer Segmentation

Segmentation is based on descriptive CLV rather than predicted CLV, realized behaviour provides a more stable and interpretable basis for tier assignment without dependence on model assumptions about future purchasing.

PLATINUM

Top 20%
\$10.69 / mo
 n = 1,702

90.7 days recency
 15.55 orders avg
 APV \$14.78

Habitual buyers at churn boundary, highest priority for proactive retention.

GOLD

Next 30%
\$3.94 / mo
 n = 2,553

171 days recency
 6.92 orders avg
 APV \$12.78

Stable mid-tier with highest upgrade potential, frequency gap vs. Platinum is the primary lever.

SILVER

Next 30%
\$1.53 / mo
 n = 2,553

274.6 days recency
 3.85 orders avg
 APV \$9.09

Occasional or seasonal buyers. Low engagement; simple win-back offers preferred over complex programs.

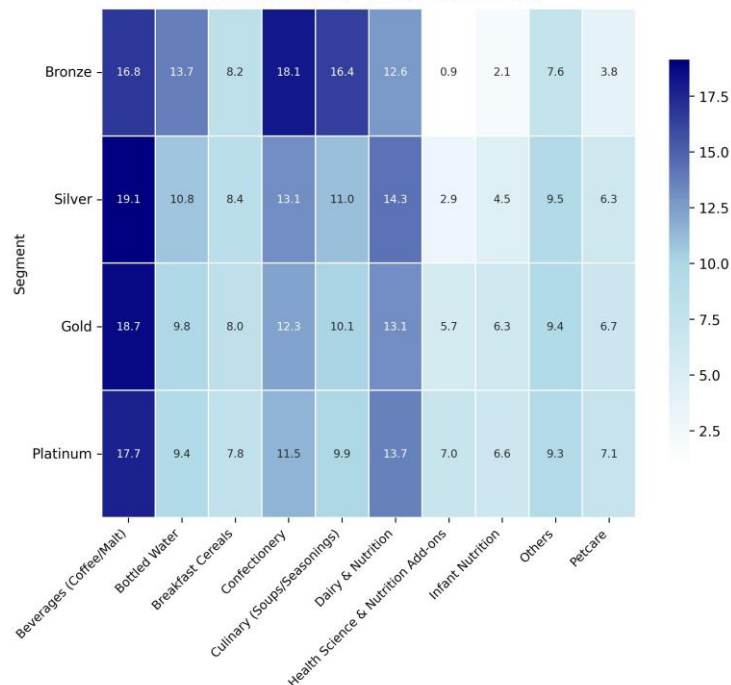
BRONZE

Bottom 20%
\$0.36 / mo
 n = 1,702

433.7 days recency
 1.77 orders avg
 APV \$4.11

Largely dormant. Near-zero CLV; minimal direct marketing investment justified on ROI grounds.

Product Category Mix by Segment (%)



6. Actionable Business Implications

1. Retention: Platinum Priority

The Platinum segment are the customers that generate the highest value. For this reason, trigger reactivation campaigns before the 90-day churn threshold must be implemented. A 1% reduction in Platinum churn has great revenue impact given their \$10.69/month CLV.

2. Loyalty Program Architecture

Tier benefits must be asymmetric: Platinum receives experiential rewards (early access, exclusive bundles) that reinforce habit without eroding margin. Gold milestones should reward incremental frequency, not spend. Bronze receives only low-cost digital win-back triggers.

3. Resource Allocation: Asymmetric budget

Top-20 predicted CLV ranges \$1,084–\$1,711. Allocate a disproportionate retention budget to Platinum and upper-Gold. Bronze customers (avg CLV \$0.36/month, recency 433 days) cannot recover acquisition cost at any reasonable payback horizon.

4. Promotion Design

Category preferences are uniform across all tiers (Beverages > Dairy > Confectionery in Platinum through Silver), so blanket category promotions will not differentially activate high-value customers. So, it's better to design promotions around frequency mechanics, not category affinity.

5. Acquisition: Quality over volume

BG/NBD confirms that once engaged, customers rarely churn. Acquisition should target behavioural profiles matching Platinum and Gold: 6–7 repeat transactions in year one is the key early indicator of high lifetime value.

Upgrade Gold → Platinum

The frequency gap (6.92 vs 15.55 orders) is the primary differentiator. Churn often affect the customers with low frequency. So, it's better to deploy replenishment reminders and subscription prompts in Beverages, Dairy, and Confectionery, the top categories across all tiers.

Thank you

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